

Sec 5.4 – Theorems about Definite Integrals

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Theorem (Properties of Limits of Integration)

If a , b , and c are any numbers and f is a continuous function, then

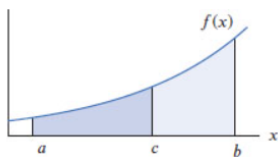
① $\int_b^a f(x) dx = - \int_a^b f(x) dx$

The integral from b to a is the negative of the integral from a to b .

② $\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$

The integral from a to c plus the integral from c to b is the integral from a to b .

$\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$ since the area under f from a to c plus the area under f from c to b together make up the whole area under f from a to b .



Partner up

Example: Given that $\int_1^5 f(x) dx = .73$ and $\int_1^2 f(x) dx = -.45$, find the value each of the following integrals.

(1) $\int_5^1 f(x) dx$

(2) $\int_2^5 f(x) dx$

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(1) $\int_5^1 f(x) dx = - \int_1^5 f(x) dx = -.73$

(2) $\int_1^2 f(x) dx + \int_2^5 f(x) dx = \int_1^5 f(x) dx.$

By substitution, $-.45 + \int_2^5 f(x) dx = .73$, so $\int_2^5 f(x) dx = 1.18$.

Note that the value got bigger, even though it is on a sub-interval. This makes sense because the first part of the interval, $[1, 2]$, had a negative integral and must have had area below the x -axis. Therefore, the positive area on the interval $[2, 5]$ must have been bigger than $.73$ to cancel out the negative area and still give the correct total.

Theorem (Properties of Sums and Constant Multiples of the Integrand)

Let f and g be continuous functions and let c be a constant.

$$\textcircled{1} \int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

The integral of the sum (or difference) of two functions is the sum (or difference) of their integrals.

$$\textcircled{2} \int_a^b cf(x) dx = c \int_a^b f(x) dx$$

The integral of a constant times a function is that constant times the integral of the function.

Partner up

Given that $\int_2^4 f(t) dt = 3$, evaluate $\int_2^4 -3f(t) + t dt$.

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By the properties,

$$\int_2^4 -3f(t) + t dt = \int_2^4 -3f(t) + \int_2^4 t dt = -3 \int_2^4 f(t) + \int_2^4 t dt.$$

We know $\int_2^4 f(t) dt = 3$.

Since $F(t) = \frac{t^2}{2}$ has a derivative of t , the FTC tells us that

$$\int_2^4 t dt = F(4) - F(2) = \frac{4^2}{2} - \frac{2^2}{2} = 8 - 2 = 6.$$

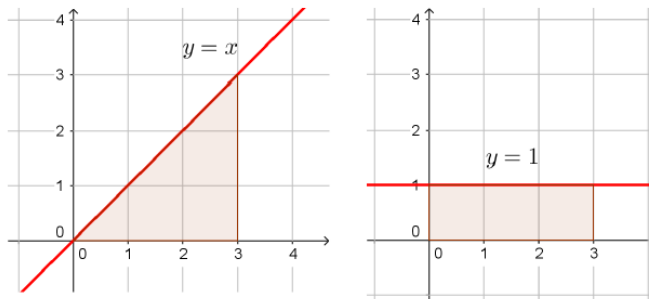
Therefore, $\int_2^4 -3f(t) - t dt = -3(3) + 6 = -3$.

This also gives another approach to evaluating the definite integral

$\int_0^3 2x - 3 \, dx$ exactly. By the sum and constant multiple properties,

$$\int_0^3 2x - 3 \, dx = \int_0^3 2x \, dx - \int_0^3 3 \, dx = 2 \int_0^3 x \, dx - 3 \int_0^3 1 \, dx$$

Both these integrals can be evaluated exactly from their graphs:



$$\int_0^3 x \, dx = \frac{9}{2} \quad \int_0^3 1 \, dx = 3$$

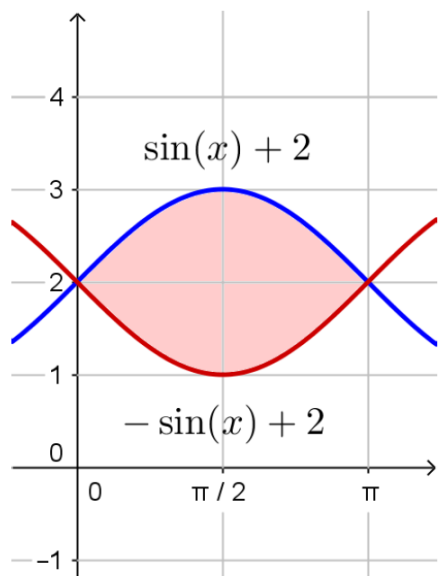
Therefore, $\int_0^3 2x - 3 \, dx = 2 \int_0^3 x \, dx - 3 \int_0^3 1 \, dx = 2 \cdot \frac{9}{2} - 3 \cdot 3 = 0.$

We can also calculate the area between two different curves:

$$\text{Area between } f \text{ and } g \text{ for } a \leq x \leq b = \int_a^b (f(x) - g(x)) \, dx.$$

Partner up

Use the sum/difference property to find the area of one section between the curves $f(x) = \sin(x) + 2$ and $g(x) = -\sin(x) + 2$:



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The two curves meet at $x = 0$ and $x = \pi$. (You could solve $\sin(x) + 2 = -\sin(x) + 2$ if you can't see this graphically)

We could approximate the height between the two curves with rectangles, where the heights from subtracting values of $g(x) = -\sin(x) + 2$ from values of $f(x) = \sin(x) + 2$. We could make this exact by taking the limit as the number of rectangles goes to infinity - which gives us the definite integral!

Therefore, we want to evaluate the definite integral

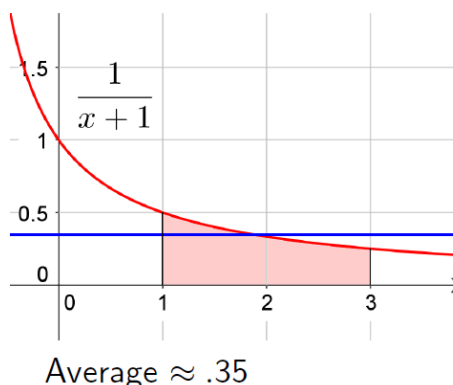
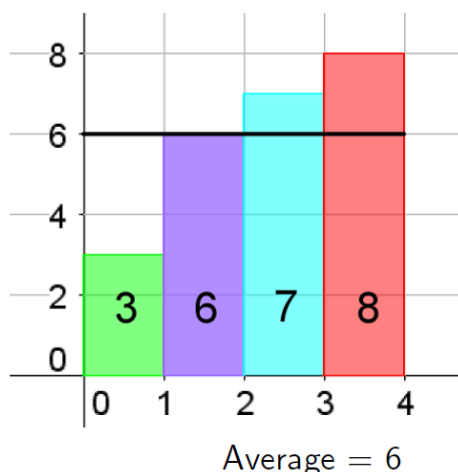
$$\int_0^{\pi} f(x) - g(x) dx = \int_0^{\pi} (\sin(x) + 2) - (-\sin(x) + 2) dx = \int_0^{\pi} 2\sin(x) dx$$

We can evaluate this using Maple or a calculator, or notice that $2\sin(x)$ is the derivative of $-2\cos(x)$ and use the FTC:

$$\int_0^{\pi} 2\sin(x) dx = -2\cos(\pi) - (-2\cos(0)) = -2(-1) + 2(1) = 4$$

Two questions:

- ① What is the average of the numbers 3, 6, 7, and 8? Can you draw a picture to *show* the average?
- ② What is the average of the function $\frac{1}{x+1}$ on the interval $[1, 3]$? Can you draw a picture to *show* the average?



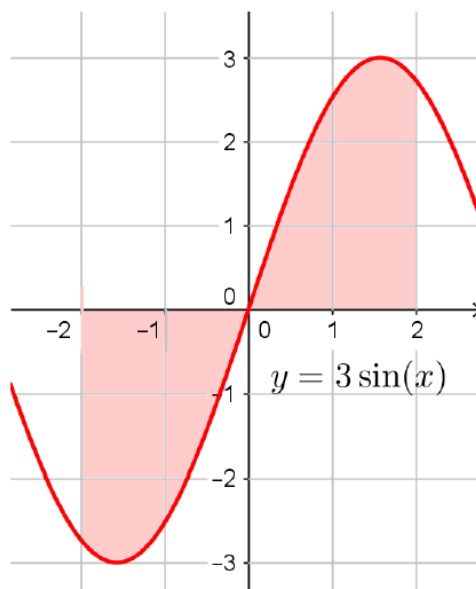
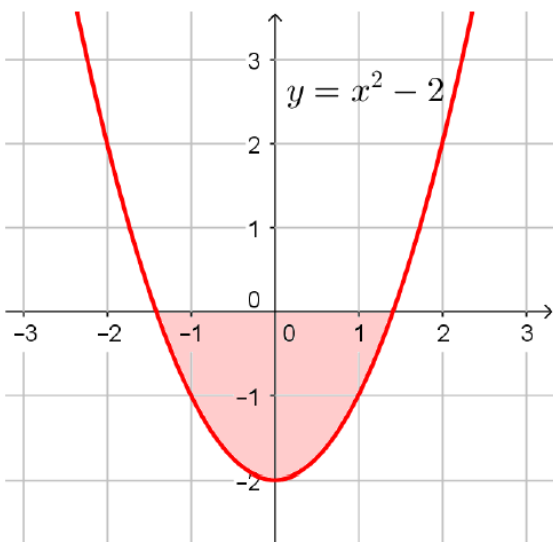
Which leads us to the following definition:

$$\text{Average value of } f \text{ from } a \text{ to } b = \frac{1}{b-a} \int_a^b f(x) dx$$

Then to find the average value of $\frac{1}{x+1}$ on $[1, 3]$, we find

$$\frac{1}{3-1} \int_1^3 \frac{1}{x+1} dx \approx .35.$$

We can use the symmetry of a function to minimize computation for integrals. What can you say about $\int_{-2}^2 f(x) dx$ for each of the following functions?



On the left, the function is symmetric about the y-axis, so $\int_{-2}^2 f(x) dx$ is double $\int_0^2 f(x) dx$ by symmetry.

On the right, the function is symmetric about the origin, so there is the same area below the x-axis as above the axis and $\int_{-2}^2 f(x) dx = 0$.

This gives examples of the following theorem:

Theorem

An **even** function is symmetric about the y-axis. Therefore,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

An **odd** function is symmetric about the origin. Therefore $\int_{-a}^a f(x) dx = 0$.